

Gradient Descent

repeat {

$$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m [(f_{w,b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)}] + \frac{\lambda}{m} w_j \right]$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(\vec{x}^{(i)}) - y^{(i)})$$

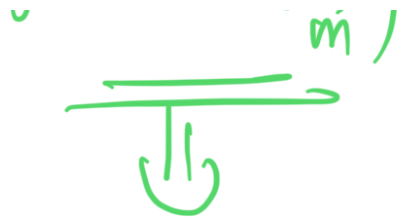
} simultaneous update.

Intuition

If we expand w_j

$$w_j = 1 \cdot w_j - \underbrace{\alpha \frac{\lambda}{m} w_j}_{w_j (1 - \alpha \frac{\lambda}{m})} - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(\vec{x}^{(i)}) - y^{(i)}) \cdot x_j^{(i)}$$

$\Downarrow$
normal



gradient
descent.

$$\text{if } \begin{matrix} \alpha = 0.01 \\ \lambda = 1 \\ m = 50 \end{matrix} \Rightarrow \alpha \frac{\lambda}{m} \approx 0.0002$$

$$\Rightarrow w_j (1 - 0.0002)$$

$$= \underbrace{0.9998}_{\text{}} w_j$$

Hence, decreasing the
value of w a little
bit per iteration.

=====